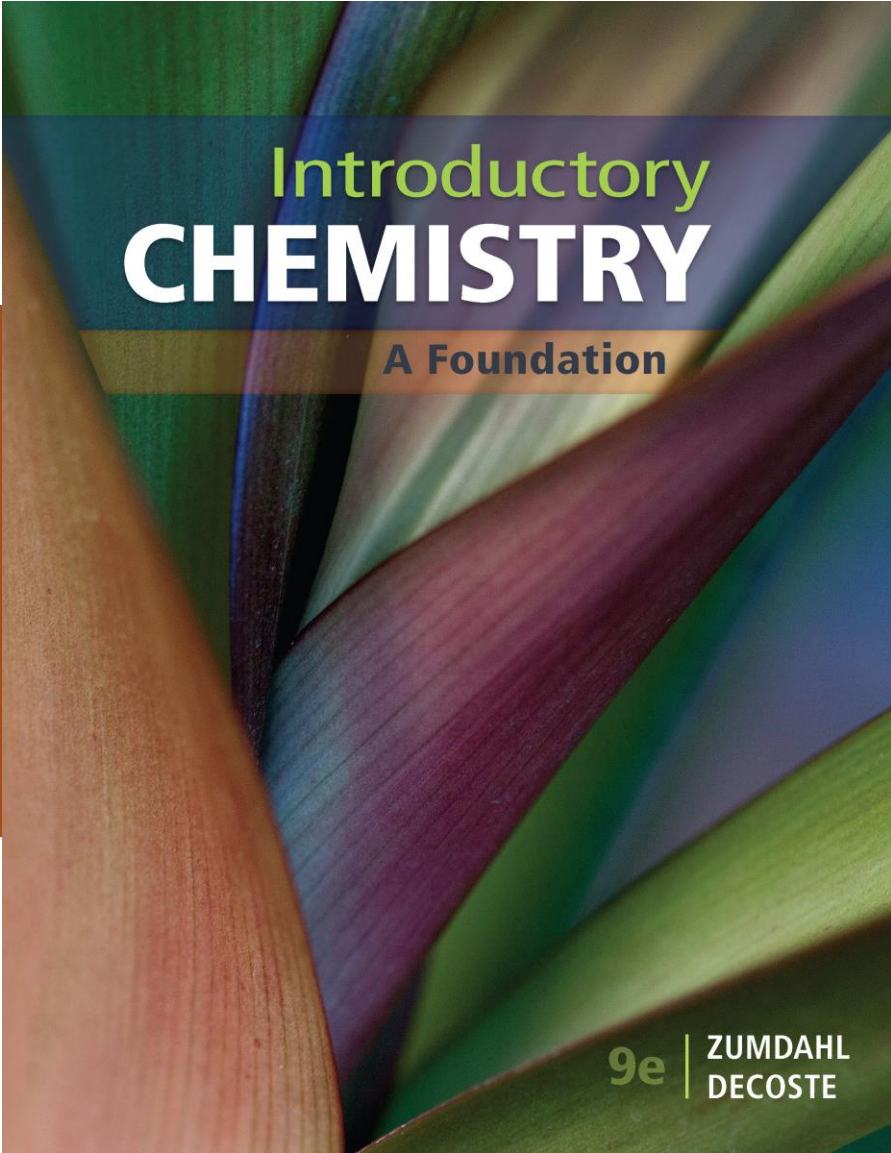




Introductory **CHEMISTRY**

A Foundation

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Chapter 5

Nomenclature

Naming Compounds

Binary Compounds

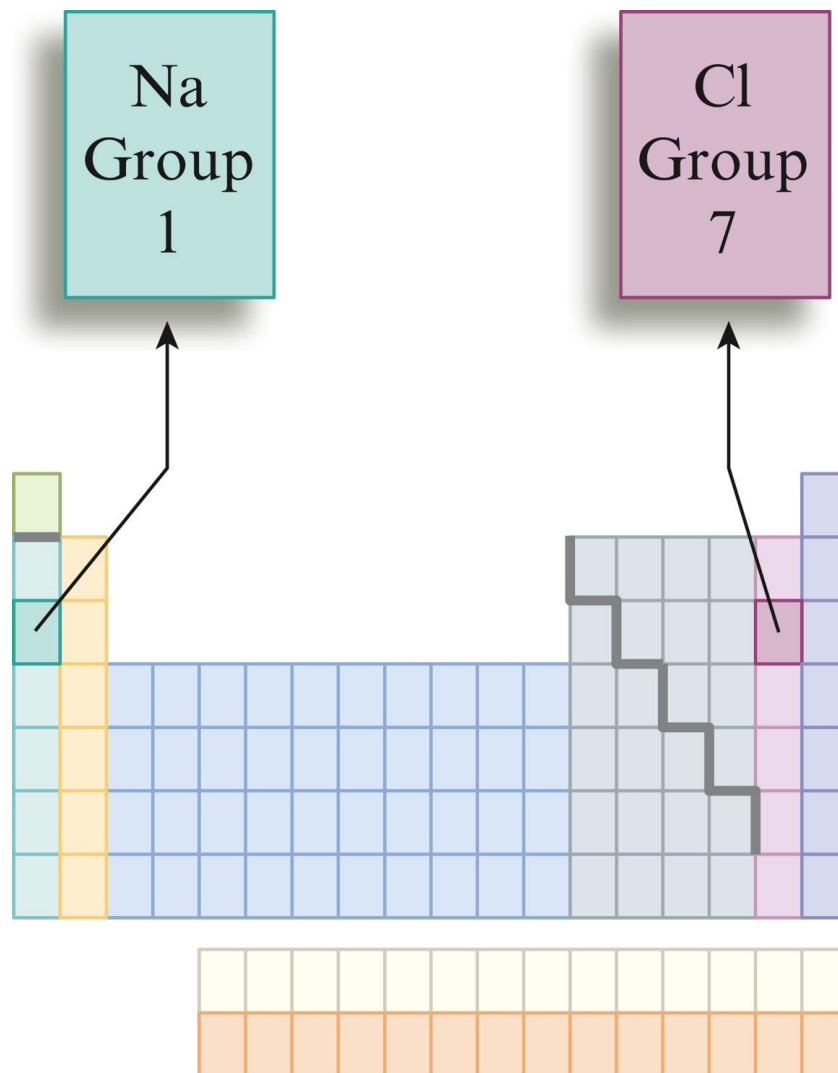
- Composed of two elements
- Divided into broad classes
 - Compounds that contain metal and a nonmetal
 - Compounds that contain two nonmetals

Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Binary Ionic Compounds

- Contain positive cations and negative anions
 - **Type I compounds**
 - Metal present forms only one cation
 - **Type II compounds**
 - Metal present can form two or more cations with different charges



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Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Common Simple Cations and Anions

Table 5.1 ▶ Common Simple Cations and Anions

Cation	Name	Anion	Name*
H ⁺	hydrogen	H ⁻	hydride
Li ⁺	lithium	F ⁻	fluoride
Na ⁺	sodium	Cl ⁻	chloride
K ⁺	potassium	Br ⁻	bromide
Cs ⁺	cesium	I ⁻	iodide
Be ²⁺	beryllium	O ²⁻	oxide
Mg ²⁺	magnesium	S ²⁻	sulfide
Ca ²⁺	calcium		
Ba ²⁺	barium		
Al ³⁺	aluminum		
Ag ⁺	silver		
Zn ²⁺	zinc		

*The root is given in color.

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Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Rules for Naming Type I Ionic Compounds

- The cation is always named first and the anion second
- A simple cation takes its name from the name of the element
- A simple anion is named by taking the first part of the element name (the root) and adding *-ide*
 - Thus the Cl^- ion is called chloride

Compound	Ions Present	Ion Names	Comments
AlCl_3	Al^{3+}	aluminum	Al (Group 3) always forms Al^{3+} .
	Cl^-	chloride	Cl (Group 7) always forms Cl^- .

Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Binary Ionic Compounds (Type I)

- Examples:

NaCl

KI

CaS

Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Exercise, 1

What is the **name** of the compound SrBr_2 ?

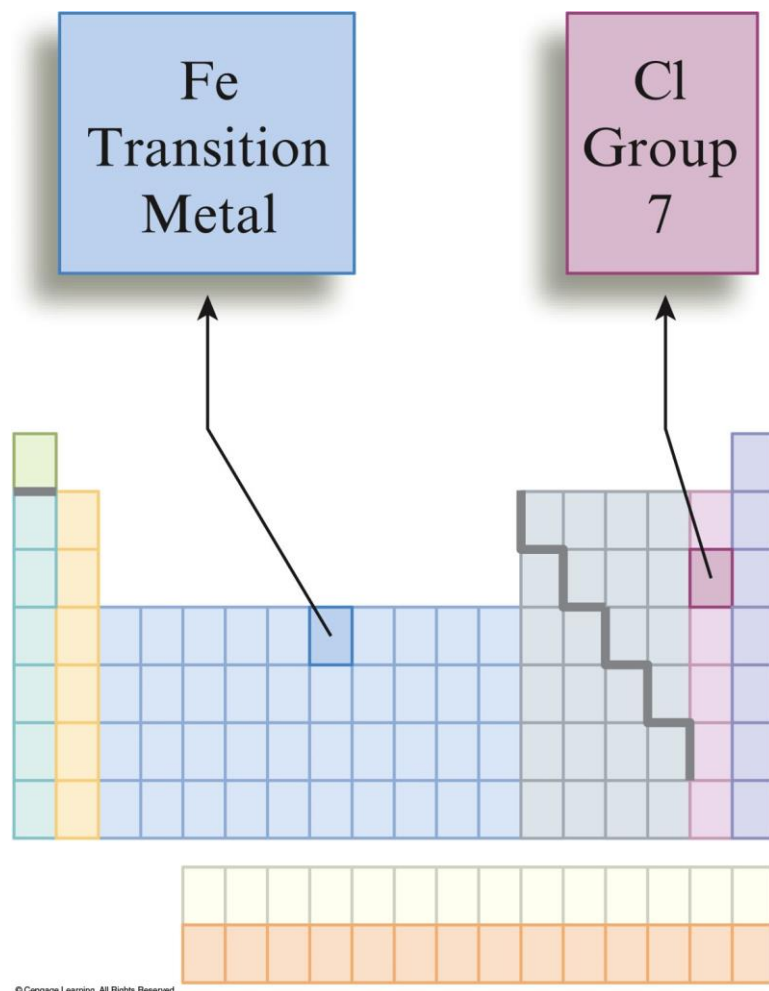
- a) Strontium bromine
- b) Sulfur bromide
- c) Strontium dibromide
- d) Strontium bromide

Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Binary Ionic Compounds (Type II)

- Metals in these compounds can form more than one type of cation
- Charge on the metal ion must be specified
- Roman numeral indicates the charge of the metal cation
- Transition metal cations usually require a Roman numeral



Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Common Type II Cations

Table 5.2 ▶ Common Type II Cations

Ion	Systematic Name	Older Name
Fe^{3+}	iron(III)	ferric
Fe^{2+}	iron(II)	ferrous
Cu^{2+}	copper(II)	cupric
Cu^{+}	copper(I)	cuprous
Co^{3+}	cobalt(III)	cobaltic
Co^{2+}	cobalt(II)	cobaltous
Sn^{4+}	tin(IV)	stannic
Sn^{2+}	tin(II)	stannous
Pb^{4+}	lead(IV)	plumbic
Pb^{2+}	lead(II)	plumbous
Hg^{2+}	mercury(II)	mercuric
Hg_2^{2+*}	mercury(I)	mercurous

*Mercury(I) ions always occur bound together in pairs to form Hg_2^{2+} .

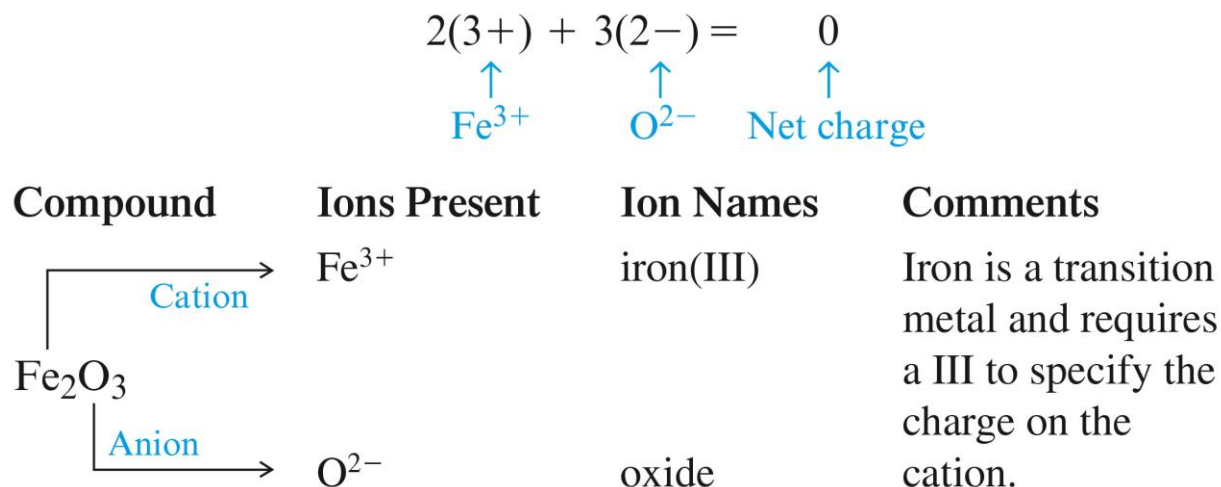
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Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Rules for Naming Type II Ionic Compounds

- The cation is always named first and the anion second
- Because the cation can assume more than one charge, the charge is specified by a Roman numeral in parentheses



Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Binary Ionic Compounds (Type II) (continued)

- Examples:

CuBr

FeS

PbO₂

Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Exercise, 2

What is the **name** of the compound CrO_2 ?

- a) Chromium oxide
- b) Chromium(ii) oxide
- c) Chromium(iv) oxide

Section 5.2

Naming Binary Compounds That Contain a Metal and a Nonmetal (Types I and II)

Exercise, 3

What is the **correct name** of the compound that results from the most stable ion for sulfur and the metal ion that contains 24 electrons?

- a) Iron(iii) sulfide
- b) Chromium(ii) sulfide
- c) Nickel(III) sulfate
- d) Iron(II) sulfide

Rules for Naming Type III Binary Compounds

- Formed between two nonmetals
 - The first element in the formula is named first, and the full element name is used
 - The second element is named as though it were an anion
 - Prefixes are used to denote the numbers of atoms present
 - The prefix mono- is never used for naming the first element

Section 5.3

Naming Binary Compounds That Contain Only Nonmetals (Type III)

Table 5.3: Prefixes Used to Indicate Numbers in Chemical Names

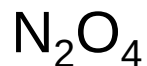
Prefix	Number Indicated
<i>mono-</i>	1
<i>di-</i>	2
<i>tri-</i>	3
<i>tetra-</i>	4
<i>penta-</i>	5
<i>hexa-</i>	6
<i>hepta-</i>	7
<i>octa-</i>	8

Section 5.3

Naming Binary Compounds That Contain Only Nonmetals (Type III)

Binary Covalent Compounds (Type III)

- Examples



Section 5.3

Naming Binary Compounds That Contain Only Nonmetals (Type III)

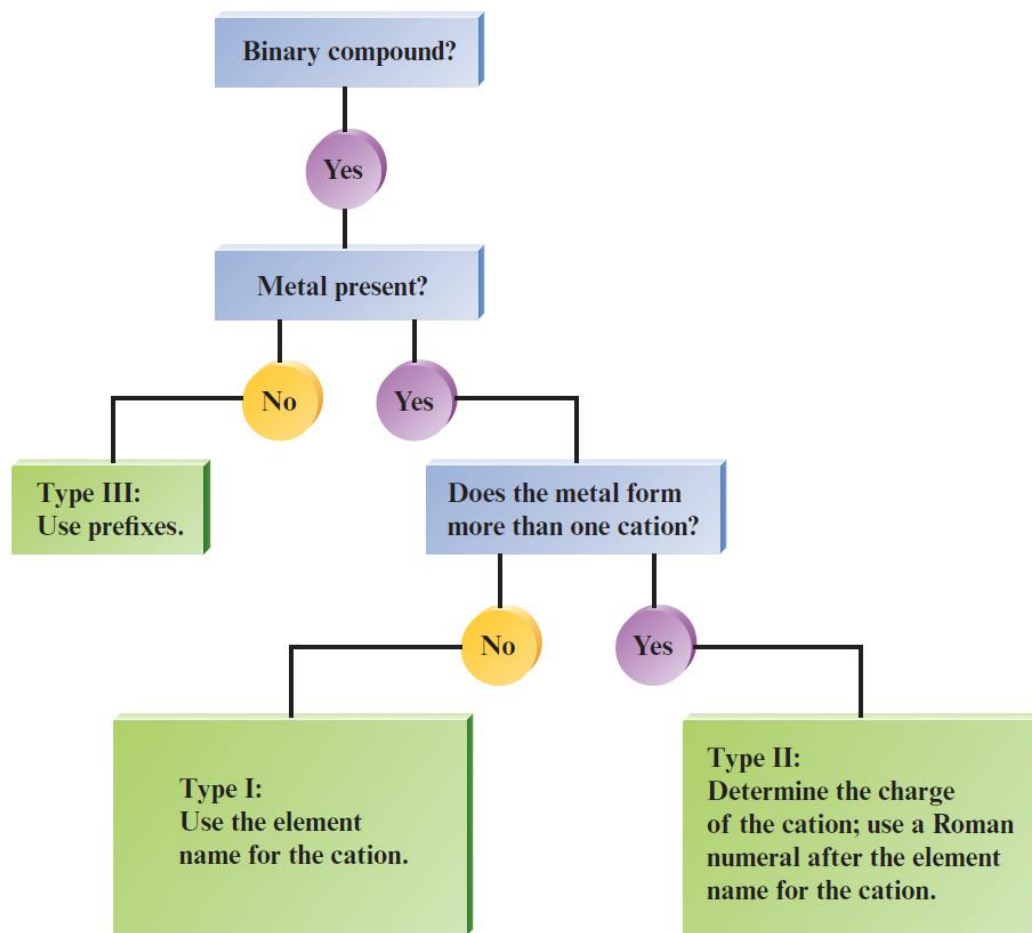
Exercise, 4

What is the **name** of the compound SeO_2 ?

- a) Selenium oxide
- b) Selenium dioxide
- c) Selenium(ii) oxide
- d) Selenium(iv) dioxide

Naming Binary Compounds: A Review

Flow Chart for Naming Binary Compounds



Naming Compounds That Contain Polyatomic Ions

Polyatomic Ions and Oxyanions

- Polyatomic ions
 - Charged entities composed of several atoms bound together
 - Have special names and must be memorized
- Oxyanions
 - Polyatomic anions exist that contain an atom of a given element and different numbers of oxygen atoms

Section 5.5

Naming Compounds That Contain Polyatomic Ions

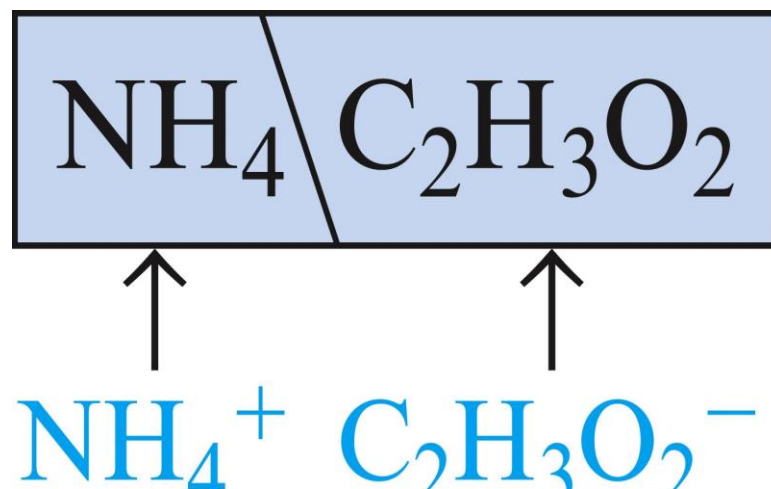
Table 5.4: Names of Common Polyatomic Ions

Ion		Name	
NH_4^+	ammonium	CO_3^{2-}	carbonate
NO_2^-	nitrite	HCO_3^-	hydrogen carbonate (bicarbonate is a widely used common name)
NO_3^-	nitrate	ClO^-	hypochlorite
SO_3^{2-}	sulfite	ClO_2^-	chlorite
SO_4^{2-}	sulfate	ClO_3^-	chlorate
HSO_4^-	hydrogen sulfate (bisulfate is a widely used common name)	ClO_4^-	perchlorate
OH^-	hydroxide	$\text{C}_2\text{H}_3\text{O}_2^-$	acetate
CN^-	cyanide	MnO_4^-	permanganate
PO_4^{3-}	phosphate	$\text{Cr}_2\text{O}_7^{2-}$	dichromate
HPO_4^{2-}	hydrogen phosphate	CrO_4^{2-}	chromate
H_2PO_4^-	dihydrogen phosphate	O_2^{2-}	peroxide

Naming Compounds That Contain Polyatomic Ions

Naming Convention

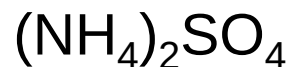
- Naming ionic compounds containing polyatomic ions follows rules similar to those for binary compounds.
 - Ammonium acetate



Naming Compounds That Contain Polyatomic Ions

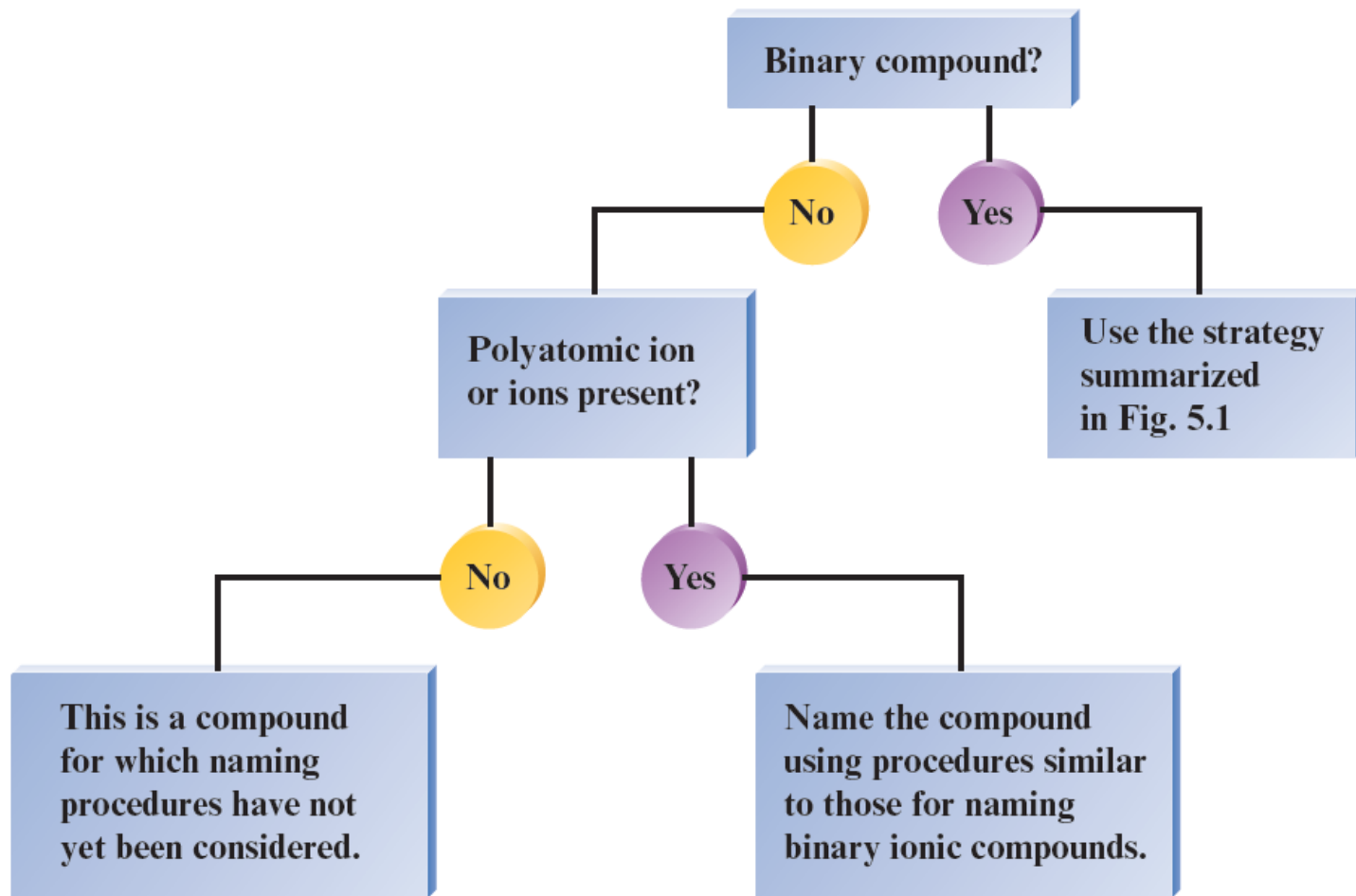
Polyatomic Ions

Examples



Naming Compounds That Contain Polyatomic Ions

Overall Strategy for Naming Chemical Compounds



Naming Compounds That Contain Polyatomic Ions

Exercise, 5

What is the **name** of the compound KClO_3 ?

- a) Potassium chlorite
- b) Potassium Chlorate
- c) Potassium Perchlorate
- d) Potassium Carbonate

Naming Compounds That Contain Polyatomic Ions

Exercise, 6

Examine the following table of formulas and names. Which of the compounds are named **correctly**?

- a) I, II
- b) I, III, IV
- c) I, IV
- d) I only**

	<i>Formula</i>	<i>Name</i>
I	P_2O_5	Diphosphorus pentoxide
II	ClO_2	Chlorine oxide
III	PbI_4	Lead iodide
IV	CuSO_4	Copper(I) sulfate

Naming Acids

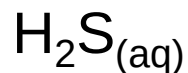
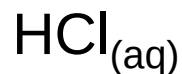
Acids

- Certain molecules produce H^+ ions when dissolved in water
- Acids can be recognized by the hydrogen that appears first in the formula— HCl
- Molecule with one or more H^+ ions attached to an anion

Naming Acids

Rules for Naming Acids

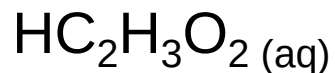
- If the anion does not contain oxygen, the acid is named with the prefix hydro– and the suffix –ic attached to the root name for the element
- Examples



Naming Acids

Rules for Naming Acids (continued)

- When the *anion contains oxygen*, the acid name is formed from the root name of the central element of the anion or the anion name, with a suffix of *-ic* or *-ous*
- Examples



Naming Acids

Table 5.5: Names of Acids That Do Not Contain Oxygen

Acid	Name
HF	hydrofluoric acid
HCl	hydrochloric acid
HBr	hydrobromic acid
HI	hydroiodic acid
HCN	hydrocyanic acid
H ₂ S	hydrosulfuric acid

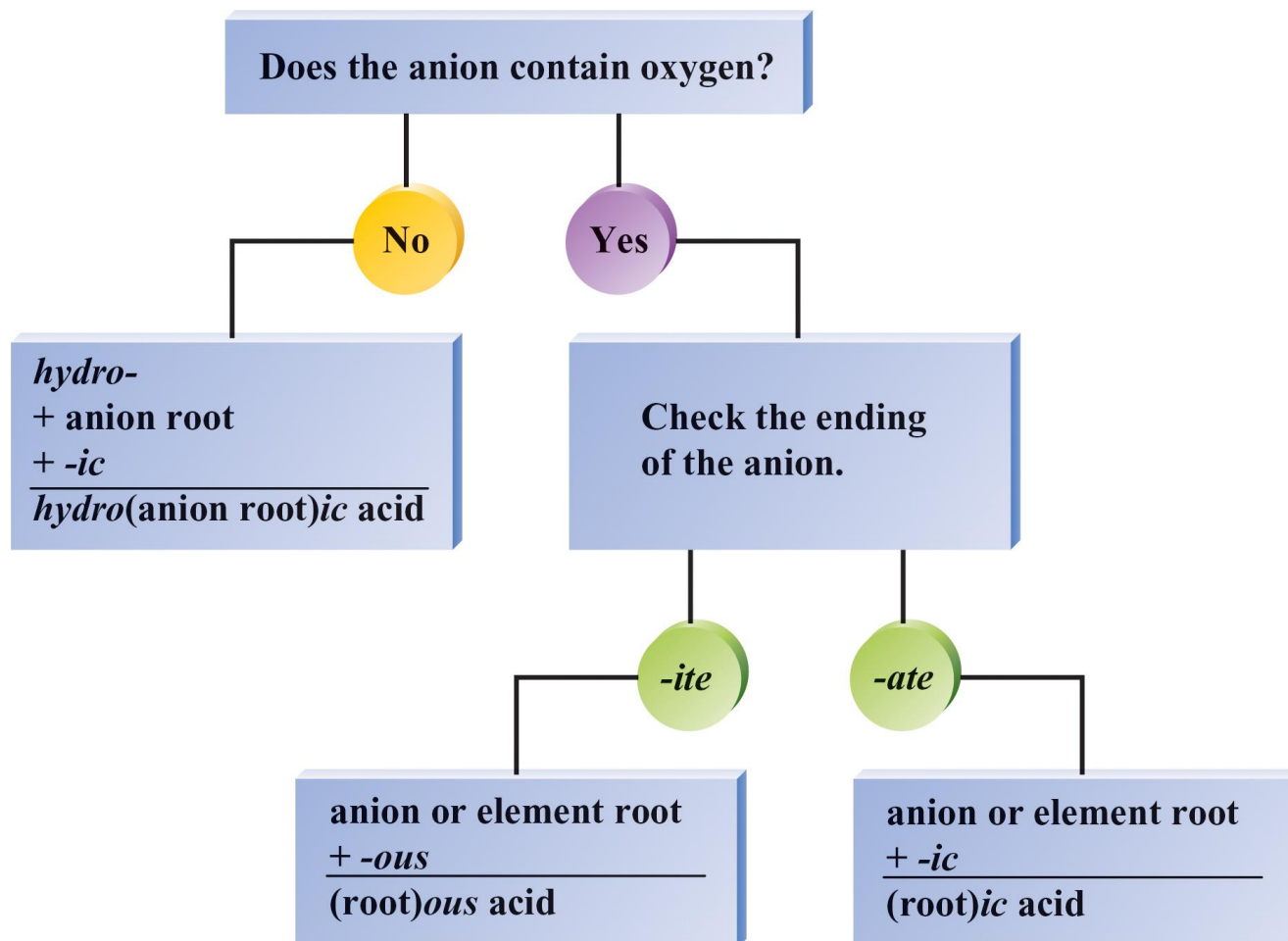
Naming Acids

Table 5.6: Names of Some Oxygen-Containing Acids

Acid	Name
HNO_3	nitric acid
HNO_2	nitrous acid
H_2SO_4	sulfuric acid
H_2SO_3	sulfurous acid
H_3PO_4	phosphoric acid
$\text{HC}_2\text{H}_3\text{O}_2$	acetic acid

Naming Acids

Figure 5.3: Flowchart for Naming Acids



Naming Acids

Exercise, 7

Which of the following compounds is named **incorrectly**?

- | | |
|-----------------------------|-------------------------|
| a) KNO_3 | Potassium nitrate |
| b) TiO_2 | Titanium(II) oxide |
| c) $\text{Sn}(\text{OH})_4$ | Tin(IV) hydroxide |
| d) PBr_5 | Phosphorus pentabromide |
| e) H_2SO_3 | Sulfurous acid |

Naming Acids

Examples, 2

- Sodium hydroxide
 - NaOH
- Potassium carbonate
 - K_2CO_3
- Sulfuric acid
 - H_2SO_4 (aq)
- Dinitrogen pentoxide
 - N_2O_5
- Cobalt(III) nitrate
 - $Co(NO_3)_3$

Naming Acids

Exercise, 8

A compound has the formula XCl_3 where X could represent a metal or nonmetal. What could the **name** of this compound be?

- a) Phosphorus trichloride
- b) Carbon monochloride
- c) Tin(iv) chloride
- d) Magnesium chloride

Writing Formulas from Names

Learning to Write the Formula of a Compound

- Essential to know the formula of a compound from its name
- Important to learn the name, composition, and charge of each of the common polyatomic anions (and the NH_4^+ cation)
- Must learn the names of common acids